

BBM/Creative Media Works: Cuts infrastructure costs by 30% while powering innovation with Google Cloud

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ABOUT BBM/Creative



With Google Cloud, Google Professional Services Organization, Creative Media Works migrated BlackBerry Messenger for Android to iOS to a cost-effective, highly available, and responsive cloud

About BBM/Creative Media Works

Creative Media Works is a division of Emtek, Indonesia's largest media, content, and social networking business. The organization operates the global BBM consumer messaging and social networking platform under license from BlackBerry Limited. BBM has evolved from a text and video messaging app into a social ecosystem that unifies chat, social, commerce, content, and services such as news, games, video, bill and top-up payment, comics, shopping, travel, career, transport, coupons, and sports.

infrastructure that
tens of millions of

Industries: Technology

Location: Indonesia

Google Cloud Results

- Delivers low latency for a global user community
- Captures and visualizes 1 petabyte of log data per month
- Manages millions of requests per second

Reduce
infrastructure
costs by
messaging
social

(<https://cloud.google.com/interconnect/>),

Pythian
love your data®



About Pythian and CloudCover

A Google Cloud Premier Partner with specializations in cloud migration, infrastructure, data analytics, and machine learning, Pythian provides IT consulting, managed services, and solutions for companies building critical data or cloud solutions using Google Cloud Platform.

A Google Cloud Premier Partner, CloudCover builds, automates, and manages systems at cloud scale.

CLOUDCOVER

Google Cloud (<https://cloud.google.com/>)

Cloud Interconnect

(<https://cloud.google.com/interconnect/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Mobile messaging providers carry their laurels. Changing consumer behavior and intense competition makes innovation a top awareness of emerging trends e

Owners and licensees of the BB responding to these drivers by e

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video messaging app into a social chat, social, commerce, games,

Cloud Dataproc

(<https://cloud.google.com/dataproc/>)

BBM is one of the largest mobile use and has attracted a dedicated particularly in Indonesia. Users of app, which was launched in 200 million times.

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud SQL (<https://cloud.google.com/sql/>)

Compute Engine

(<https://cloud.google.com/compute/>)

Emtek signs BBM licensing agreement

To build its partner network and continue to innovate in content, commerce (online and online to offline), and finance, PT Elang Mahkota Teknologi Tbk (Emtek), one of Indonesia's largest media, content, and technology businesses, signed a licensing agreement in 2016 with BlackBerry Limited. Under this agreement, Emtek division Creative Media Works operates the global **BBM** (<https://www.bbm.com/en/about.html>) consumer messaging and social networking platform.

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Creative Media Works now provides BBM Android and iOS features, as well as services and content, to the global market. BlackBerry retains control of the BBOS and BlackBerry 10 versions of BBM, and BBM Enterprise.

Creative Media Works now operates as a standalone business with teams in Canada, Singapore, Indonesia, the United Arab Emirates, and South Africa. "The licensing agreement incorporated a hosting arrangement with BlackBerry for up to two years, giving us time to identify a vendor to support our development work and cut over to them," explains Mohan Krishnan, Chief Technology Officer, BBM.

"To help us with this task, the BlackBerry team provided a bill of materials specifying the components comprising the environment BBM needs," adds Krishnan. "From that bill of materials, we undertook a discovery exercise to determine whether the product is running on physical servers or virtual machines. We also looked at what

technology stacks are involved, how those stacks are deployed and managed, and how best to migrate from the BlackBerry data center to a new provider."

The business quickly determined that a hosted cloud environment should meet its agility, speed, and cost requirements.

Creative Media Works selects Google

Creative Media Works shortlisted three cloud vendors and conducted two-week proofs of concept with each. "We benchmarked and mapped each service against the BlackBerry bill of materials and calculated an estimated hosting cost over three to five years," says Krishnan. "We selected Google Cloud Platform (<https://cloud.google.com/>) because it provided the widest range of high-quality technology and services, at a competitive cost."

The business was particularly interested in the potential of Google Cloud Platform custom virtual machine instances to optimize its compute usage; its expertise in database management; and its logging and monitoring.

Creative Media Works then engaged Google Cloud Premier Partner and Specialization Partner Pythian (<https://pythian.com/>) to help deliver the project. "We chose Pythian because of the type of expertise it provides, especially in the data space," Krishnan says. "The partner has people with the technical skills to get into the trenches and solve tough

problems. That was a characteristic we felt was really critical for us to be successful."

"We also chose CloudCover (<https://cldcvr.com/>) to assist with the migration as it was a member of the Google Premier Partner network, with extensive experience working with Google Cloud," he adds.

India-based CloudCover operated in a time zone that was user friendly for the BBM team in Indonesia.

"CloudCover representatives came on site to the BBM premises in Jakarta several times over a four to six week period, and worked closely with our various teams during the migration project," Krishnan says.

With assistance from the Google Professional Services (<https://cloud.google.com/consulting/>) team, CloudCover, and Pythian, Creative Media Works embarked on a project to migrate BBM for Android, iOS, and Windows into Google Cloud Platform data centers in Singapore and Taiwan. The business used Cloud Interconnect (<https://cloud.google.com/hybrid-connectivity/>) – a Google networking service – to link the BlackBerry data centers in North America to its data centers in Asia.

"Cloud Interconnect enabled us to use dedicated fiber-optic links to connect from the Google network to the BlackBerry infrastructure, allowing us to control our destiny and assure high performance," says Yu-Wern Pong, Project Manager, BBM at Creative Media Works. "This enabled us to phase the migration of our components over more than 14,000

kilometers from the BlackBerry data center to our Google Cloud Platform environments."

Pythian provided data architects and solution engineers to help design the project and provide technical guidance. This helped ensure the architecture provided the required flexibility and scalability for the BBM app.

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Professional Services overcomes challenges

The Google Professional Services Organization assigned two Technical Account Managers – one in the Asia-Pacific and one in North America – to help ensure Creative Media Works had assistance available when required. The Professional Services Organization team also facilitated several consulting workshops to build Creative Media Works' knowledge of Google Cloud Platform architecture and specific products such as BigQuery (<https://cloud.google.com/bigquery/>) and Cloud Pub/Sub (<https://cloud.google.com/pubsub/>).

"The Google Professional Services Organization worked closely with our migration team and used its deep knowledge of products and best practices to help us overcome any challenges," Krishnan explains. "The Professional Services team was instrumental in facilitating discussions with Google product managers and technical specialists to help us use Cloud Interconnect and Google managed services effectively."

"The Google Technical Account Managers were particularly helpful in designing and implementing a logging and data analytics infrastructure," he adds.

Google Professional Services Organization's support over a one-year engagement enabled Creative Media Works to complete the project early last year.

The migration is delivering immediate benefits to Creative Media Works. The organization now runs

BBM for iOS and Android in a highly efficient Google Cloud Platform architecture.

The global presence of Google Cloud Platform and its load balancing service has enabled the business to centralize its infrastructure in select regions while continuing to provide a low-latency experience for the global BBM subscriber base.

Users' network traffic is ingested at the closest Google point of presence and traverses the Google backbone to virtual machine instances run by BBM.

The Cloud Pub/Sub messaging service gathers and forwards all events to the Creative Media Works ecosystem, while Cloud Dataflow (<https://cloud.google.com/dataflow/>) enables the business to rely on a single cloud-based managed service for batch and stream processing.

The business uses the two services to feed log data into an Elasticsearch cluster for near-real-time visualization and real-time analysis. The same pipeline feeds into its downstream data analytics process and into a BigQuery analytics data warehouse for business intelligence reporting.

Meanwhile, the combination of BigQuery and the Cloud Dataproc (<https://cloud.google.com/dataproc/>) service for cluster management enables the Creative Media Works team to obtain answers to complex queries in a couple of minutes rather than hours.

This allows the business to perform more frequent analysis to guide product development, feature testing, and the creation of more intelligent user-facing features.

Meanwhile, Compute Engine

(<https://cloud.google.com/compute/>) provides infrastructure-as-a-service that manages millions of requests per second, while Cloud Pub/Sub provides messaging to support about 1 petabyte of log data every month.

Google virtual machine instances have enabled Creative Media Works to optimize the cost and scalability of its infrastructure. "Google's support for custom virtual machine instances allowed us to determine our preferred CPU-to-memory ratio, allowing us to maximize utilization based on the consumption model for our backend services," explains Yu-Wern.

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"Some of these services may be CPU-bound and others memory-bound. Google Cloud enables us to spin up big virtual machines – such as a 64 vCPU node with 240GB RAM – which we could only previously achieve on bare-metal nodes. This allows us to scale vertically, reducing our node count and operational overhead."

With custom machine instances, Creative Media Works has reduced the number of virtual machine instances required for core messaging from about

5,000 on the BlackBerry data center infrastructure to about 3,500 on Google Cloud Platform, with autoscaling adjusting virtual machine capacity as required.

For media storage, Creative Media Works has migrated from about 500 instances, NetApp filers, and solid-state disk over FusionIO to pay-per-use on Cloud Storage (<https://cloud.google.com/storage/>).

For logging and monitoring, the business has migrated from about 5,000 instances to Cloud Pub/Sub and Cloud Dataflow.

Creative Media Works also moved from about 1,000 instances used for big data and analytics to Dataproc and BigQuery, and 500 instances running its databases, to the Cloud SQL (<https://cloud.google.com/sql/>) managed service for relational databases.

The business has also taken advantage of the Google Committed Use Discount (CUD) service – that allows it to purchase a contract for one year or three years in return for deeply discounted pricing for vCPUs and memory – to further control its infrastructure costs.

Managed services key

Google Cloud Platform managed services such as Cloud Dataproc have also played a key role in reducing operational complexities and overheads.

These services help ensure Creative Media Works does not have to manage software upgrades – and can simply access the latest versions of the Apache Hadoop big data processing framework or the Apache Spark analytics engine as required.

Furthermore, when the business has to run a job, it need only spin up an Apache Hadoop cluster on Cloud Dataproc, perform the task, and collate the results before terminating the cluster. It only pays for the resources it needs during this process.

"Per-minute billing means we can complete this task with minimum waste of idle resources," says Yu-Wern. "Overall, I would estimate managed services reduced our BBM infrastructure costs by 30 percent. With Google handling the scaling of our managed services, our engineering team can focus on other key operational areas, and we do not have to recruit people with specialist skill sets."

With Google Cloud Platform, Creative Media Works is now incurring about half the infrastructure costs it would have incurred had it remained in the BlackBerry data centers, while recording near-zero downtime.

For Krishnan, Google Cloud Platform has been integral to the success of a critical and closely watched project. "As a technology company, our bread and butter is to deploy quickly in a reliable infrastructure," he explains. "Our board was tracking this particular migration, because of the opportunities it presented to reduce our costs, deploy new products and features, and build the

BBM service. With Google, we are achieving our goals."

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